

Stage 1



Role Assignment

- ▶ Balanced Good/Evil assignment
- ▶ Constraint-satisfying problem (CSP)
- ▶ Macro-distribution solver

250 games

3G / 2E

Deterministic

Stage 2



Public History

- ▶ Constraint-satisfying team composition, votes and quest outcomes
- ▶ 0 hard constraint violations

250 games

Deterministic

Suspicion scores

Stage 3



Dialogue Generation

- ▶ Two-pass pipeline:
- ▶ Precompute Tactic: Cyclic (R1) & GPT-5.2 (R2–R5)
- ▶ Generate Dialogues: GPT-5.2 & Gemini-3.1

37 tactics

2 dialogue sets

Max coverage

Per round

Stage 4



Dialogue Verification

- ▶ LLM-as-a-Judge: Claude Sonnet 4.5
- ▶ 5 criteria binary scoring
- ▶ Targeted & Custom corrections

Coherence

Authenticity

Blind A / B

2 layer verifier

Stage 5



Reasoning Traces

- ▶ ToM annotation via ReCon and ReAct frameworks
- ▶ Abductive hypothesis, suspicion scores, gated ToM layers

Cognitive map

Abduction

Suspicion

ToM $k \leq 2$

Stage 6



Trace Verification

- ▶ LLM-as-a-Judge: Claude Sonnet 4.6
- ▶ 4 criteria binary scoring
- ▶ Human-in-the-loop random verification

Soundness

ToM depth

Belief consistency